

Day – 2 Secure network with firewalls

Task – 1 Setup windows firewall defender.

- Step-by-Step Instructions
- Open Windows Defender Firewall with Advanced Security:
- Press the Windows Key, type wf.msc, and press Enter.
- Navigate to Inbound Rules:
- In the left-hand pane of the Firewall console, click on Inbound Rules to select it.
- Create a New Rule:
- In the right-hand Actions pane, click on New Rule... to launch the New Inbound Rule Wizard.
- Select Rule Type:
- Choose Port and click Next.
- Specify the Port:
- Select TCP (or UDP, depending on your requirement; standard web traffic typically uses TCP).
- Select Specific local ports and enter 8080. Click Next.
- Choose the Action:
- Select Block the connection and click Next.
- Select Profile:
- Check the boxes for the network profiles where this rule should apply (Domain, Private, and Public) and click Next.
- Name and Finish:
- Give the rule a clear name (e.g., Block Port 8080) and an optional description.
- Click Finish to save and activate the rule.

Task – 2 Simulate NAT.

- [Phone] --> Private IP: 192.168.1.10
- [Laptop] --> Private IP: 192.168.1.11 \
- [Smart TV] --> Private IP: 192.168.1.15 /---> [Wi-Fi Router (NAT)] ---> Public IP: 203.0.113.45 ---> [The Internet]
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- How NAT Works:
- Private Local Addresses: Inside your home network, your Wi-Fi router assigns unique private IP addresses (such as 192.168.1.x) to each connected device like your phone, laptop, and smart TV so they can communicate locally.

- The Translation Process: Because private IP addresses are not routable on the public internet, when your phone or laptop requests a website, the router intercepts the packet. It swaps the device's private source IP address with its own single public IP address (assigned by your ISP) and tracks the connection using port numbers.
- Returning Traffic: When data comes back from the internet, the router looks at its NAT translation table, identifies which local device made the request, replaces the destination with that device's private IP, and forwards the data to the correct gadget.

Task – 3 Config port filtering.

- To block outgoing traffic on port 25, log into your Wi-Fi router's admin panel, navigate to the Security, Firewall, or Access Control menu, and create a rule to deny outbound TCP traffic on port 25.
- Blocking outgoing port 25 stops compromised or infected local devices from directly connecting to external mail servers to blast out unsolicited bulk emails. This prevents your home network's public IP address from getting blacklisted by spam filtering services.

Task – 4 Sample packet Filtering rule set.

- Sample Rule Set
- Rule 1: Deny TCP traffic from any source to any destination on Port 22 (SSH).
- Rule 2: Permit TCP traffic from the internal network (192.168.1.0/24) to any destination on Port 80 (HTTP).
- Rule 3: Permit traffic from the internal network (192.168.1.0/24) to Port 53 (DNS).
- Rule 4: Deny all other traffic (Implicit Deny).
- Packet Analysis and Verdicts
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- 1. HTTP request from 192.168.1.5 to 172.217.166.78:80
- Evaluation: The source IP falls within the internal network range (192.168.1.x), and the destination port is 80 (HTTP). It directly matches Rule 2.
- Verdict: Allowed
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- 2. SSH request from 10.0.0.10 to 34.201.23.1:22
- Evaluation: The destination port is 22 (SSH), which triggers Rule 1 regardless of the source IP.
- Verdict: Blocked
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- 3. DNS query from 192.168.1.7 to 8.8.8.8:53
- Evaluation: The source IP is within the internal network (192.168.1.x), and the destination port is 53 (DNS), matching Rule 3.
- Verdict: Allowed

Task – 5 List three ways to protect server.

- 1. Enforcing Port and Protocol Restrictions
- At the network perimeter, packet filtering acts as the initial gatekeeper by evaluating the headers of incoming data packets against strict, predefined rules.
- Restricting Entry Points: They explicitly permit standard web traffic (such as HTTPS on port 443) while blocking all external access to non-essential ports, preventing unauthorized services from being exposed.
- Dropping Malformed Packets: Packets featuring corrupted headers, invalid TCP flags, or unrecognized transport protocols are discarded immediately before they can consume application server resources.
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- 2. Mitigating DDoS Attacks and IP-Based Misuse
- Platforms frequently face automated traffic surges, data scraping bots, and Distributed Denial of Service (DDoS) attacks. Firewalls help mitigate these threats at the network layer.
- Traffic Rate-Limiting: Stateful packet filtering tracks active connections and automatically drops packets when a single source IP address exceeds normal request thresholds.
- Blacklist and Edge Filtering: Network administrators configure rules to filter out incoming data packets originating from known malicious IP ranges, proxy pools, or blocked geographic zones.
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- 3. Microsegmentation and Internal Zoning
- Security doesn't stop at the outer boundary; large-scale architectures use internal packet filtering rules to protect sensitive assets like user profiles, direct messages, or financial transaction histories.
- Isolating Database Tiers: Internal firewalls use Access Control Lists (ACLs) to ensure that backend databases only accept packets originating from verified, authorized frontend application servers.
- Preventing Lateral Movement: If an attacker compromises a single public-facing web server, internal packet filters restrict them from freely scanning or accessing other internal microservices.
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